

Dependence, Work, and Entanglement: Corrections and Extensions to the Oscillator Notes

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Abstract

This note does two things. First, it tightens the bridge between the classical coupled-oscillator model and the Yang *et al.* membrane experiment: the experiment's own normal-mode splitting fixes $|g|/k \simeq 2|\Lambda|/\omega_0$ (not Λ/ω_0), and the experiment is a two-temperature steady state, in which the dominant dependence is a heat-current correlation 10^3 – 10^8 times larger than the equilibrium estimate. Second, it puts the classical and quantum notes on a common footing. An exact free-energy decomposition explains why decoupling an equilibrium-correlated pair costs work while unsqueezing a two-mode squeezed state releases it; the quantum Gibbs state of the same coupled oscillators recovers $W = k_B T \Delta I$ in the classical limit and shows where entanglement appears; and a squeezed thermal family shows that released work tracks neither I nor E_N . All numbers are reproduced by `extensions_calculations.py`.

1 The coupling bridge: $q = 2|\Lambda|/\omega_0$

Yang *et al.* write the mode dynamics (their Eq. 1) as $i\partial_t(b_1, b_2)^T = \begin{pmatrix} \omega_0 - i\gamma_1/2 + \Lambda & \Lambda \\ \Lambda & \omega_0 - i\gamma_2/2 + \Lambda \end{pmatrix} (b_1, b_2)^T + \text{noise}$, with normal modes at ω_0 and $\omega_0 + 2\Lambda$ ($\Lambda < 0$: the optical spring is negative).[1] For the classical model $E = \frac{k}{2}(x^2 + y^2) + gxy$ with mass m , the normal modes are $\omega_{\pm} = \omega_0 \sqrt{1 \pm q}$, $q = |g|/k$, split by $\omega_0 q + O(q^3)$. Matching splittings gives

$$q = \frac{|g|}{k} \simeq \frac{2|\Lambda|}{\omega_0}. \quad (1)$$

The same follows by writing $g x_1 x_2$ with $x_i = x_{\text{zpf}}(b_i + b_i^\dagger)$, $x_{\text{zpf}}^2 = \hbar/2m\omega_0$, and keeping resonant terms: $\hbar\Lambda = g x_{\text{zpf}}^2$, i.e. $\Lambda = g/2m\omega_0$. It is also why the heat flux oscillates at 2Λ : that is the normal-mode beat frequency. The earlier identification $q \approx \Lambda/\omega_0$ was low by a factor of 2, so $I \simeq q^2/2$ and W_{rev} were low by a factor of 4:

$\Lambda/2\pi$ (Hz)	$q = 2 \Lambda /\omega_0$	I (nats), corrected	I (nats), previous
6	3.00×10^{-5}	4.50×10^{-10}	1.13×10^{-10}
8	4.00×10^{-5}	8.00×10^{-10}	2.00×10^{-10}
58	2.90×10^{-4}	4.21×10^{-8}	1.05×10^{-8}
85	4.25×10^{-4}	9.03×10^{-8}	2.26×10^{-8}

For $85 \rightarrow 58$ Hz, $\Delta I = 4.83 \times 10^{-8}$ nats and $W_{\text{rev}} = k_B T \Delta I = 2.0 \times 10^{-28}$ J at 300 K (previously 5.0×10^{-29} J). The fractional drop in I (53.4%) is unchanged.

Two smaller corrections. (i) In Yang's Eq. 1 the amplitude decays at $\gamma_i/2$, so $1/\gamma_1 = 13.3$ ms and $1/\gamma_2 = 26.5$ ms are *energy*-decay times; amplitude-decay times are 26.5 and 53.1 ms. The conclusion that the 8.62 ms flux period is shorter still holds. (ii) The reference is *Nat. Commun.* **11**, 4656 (2020), not 5933. For context, Yang's strong-coupling threshold is $|\Lambda| = |\gamma_1 - \gamma_2|/4$, i.e. $|\Lambda|/2\pi = 1.5$ Hz, so all four reported settings are in the strong-coupling regime.

2 The experiment is a steady state, not an equilibrium

In the experiment, membrane 1 is driven by white noise (hot bath T_H) and membrane 2 sits at room temperature (T_L). The equilibrium result $W_{\text{rev}} = \Delta F$ assumes a Gibbs state, which this is not. Solving Yang's Eq. 1 in steady state with thermal occupations $n_i = k_B T_i / \hbar \omega_0$ gives, in closed form,

$$s \equiv \text{Im}\langle b_1 b_2^\dagger \rangle = \frac{2\Lambda\gamma_1\gamma_2(n_1 - n_2)}{(\gamma_1 + \gamma_2)(\gamma_1\gamma_2 + 4\Lambda^2)}, \quad N_{11} = n_1 - \frac{2\Lambda s}{\gamma_1}, \quad N_{22} = n_2 + \frac{2\Lambda s}{\gamma_2}, \quad (2)$$

with $\text{Re}\langle b_1 b_2^\dagger \rangle = 0$ on resonance. The heat current is $J = 2\hbar\omega_0\Lambda s$, which reproduces Yang's saturated value $\gamma_1\gamma_2 k_B(T_H - T_L)/(\gamma_1 + \gamma_2)$. Since the steady state is Gaussian, the mutual information of the full phase-space state $(u_1, \dot{u}_1; u_2, \dot{u}_2)$ is

$$I_{\text{ph}} = -\log(1 - |\rho|^2), \quad |\rho| = \frac{|s|}{\sqrt{N_{11}N_{22}}} = \frac{|J|}{2|\Lambda| k_B \sqrt{T_1^{\text{eff}} T_2^{\text{eff}}}}. \quad (3)$$

In the lab frame this correlation lives between u_1 and $\dot{u}_2$, a quarter period out of phase. It is exactly the quantity Yang *et al.* integrate to measure heat flux, $j \propto -\langle u_1 \dot{u}_2 \rangle$. The equal-time position correlation stays at $\simeq q$. A full non-rotating-wave Langevin solution (Lyapunov equation, no approximations; checked against the Gibbs value at $T_1 = T_2$) agrees with Eq. (3) to four digits and gives:

$\Lambda/2\pi$ (Hz)	Gibbs (Sec. 1)	$T_H/T_L = 2$		$T_H/T_L = 10$	
	I	I_{pos}	I_{ph}	I_{pos}	I_{ph}
6	4.50×10^{-10}	4.87×10^{-10}	1.94×10^{-2}	5.51×10^{-10}	1.05×10^{-1}
8	8.00×10^{-10}	8.40×10^{-10}	1.45×10^{-2}	9.01×10^{-10}	7.35×10^{-2}
58	4.21×10^{-8}	4.21×10^{-8}	4.24×10^{-4}	4.22×10^{-8}	1.95×10^{-3}
85	9.03×10^{-8}	9.04×10^{-8}	1.98×10^{-4}	9.05×10^{-8}	9.12×10^{-4}

Three consequences follow. (i) The equilibrium estimate survives only for equal-time *positions*; it is within 25% of I_{pos} at every setting. (ii) The dominant dependence is sustained by the heat current and is 10^3 – 10^8 times larger. It is *largest at weak coupling*, peaking near $4\Lambda^2 = \gamma_1\gamma_2$, and falls as Λ grows and the two effective temperatures equalize. This is the opposite trend to the equilibrium model. (iii) There is no free energy for a steady state, so $W = k_B T \Delta I$ has no direct analogue here. What maintains the correlation is a continuous entropy production $\dot{S} = J(1/T_L - 1/T_H)$, and the natural framework is information flow in steady states.[2] The practical upshot is that Eq. (3) needs only J , Λ , and $T_{1,2}^{\text{eff}}$, all of which Yang *et al.* plot (their Figs. 2d, 3b, 5a). I_{ph} can therefore be estimated at every reported setting without new measurements.

3 One identity behind both notes

For $H = H_A + H_B + V$ and any state ρ_{AB} , the nonequilibrium free energy $\mathcal{F}(\rho; H) = \text{Tr} \rho H - k_B T S(\rho)$ splits exactly as

$$\mathcal{F}(\rho_{AB}; H) = \mathcal{F}(\rho_A; H_A) + \mathcal{F}(\rho_B; H_B) + k_B T I(A:B) + \langle V \rangle, \quad (4)$$

classically and quantum mechanically, because $S_{AB} = S_A + S_B - I$. Since $\mathcal{F}(\rho) - F_{\text{eq}} = k_B T D(\rho||\gamma)$ bounds isothermal work extraction,[3] correlations are always a free-energy *resource* worth exactly $k_B T I$. [4, 5] Whether removing them costs or releases work depends on the other terms.

Classical note. For the Gibbs state at coupling q , relative to the uncoupled equilibrium:

$$F(g) - F(0) = \underbrace{k_B T(s - 1 - \ln s)}_{\text{local, } O(q^4)} + \underbrace{k_B T I}_{\simeq k_B T q^2/2} + \underbrace{\langle V \rangle}_{=-k_B T q^2/(1-q^2)} = -k_B T I, \quad s = \frac{1}{1-q^2}. \quad (5)$$

Decoupling costs work because the interaction energy $\langle V \rangle \approx -2k_B T I$ must be paid back, and the correlation free energy refunds half of it. The law $W = k_B T \Delta I$ is this specific cancellation, which is why it is model-specific.

Quantum note. For the two-mode squeezed vacuum, $V = 0$ at both endpoints and the correlations sit on top of excess local energy, so every term in Eq. (4) is released when they are removed. The sign difference between the two notes is therefore *not* classical versus quantum. It is equilibrium correlations bound by an interaction versus nonequilibrium correlations carried by excess energy.

4 Quantum equilibrium counterpart of the classical model

Quantizing the same Hamiltonian gives normal modes $\omega_{\pm} = \omega_0 \sqrt{1 \pm q}$. The Gibbs state is Gaussian, with free energy $F = \sum_{\pm} k_B T \ln[2 \sinh(\hbar\omega_{\pm}/2k_B T)]$. Partial transposition gives the entanglement condition $\coth(\hbar\omega_+/2k_B T) \coth(\hbar\omega_-/2k_B T) < \omega_+/\omega_-$ and, at $T = 0$, $E_N = \frac{1}{4} \ln \frac{1+q}{1-q} \simeq q/2$. [7] For $q = 0.2$:

$k_B T / \hbar \omega_0$	decoupling cost $-\Delta F$ ($\hbar \omega_0$)	$-\Delta F / (k_B T I)$	E_N (nats)
$\rightarrow 0$	5.06×10^{-3}	$\rightarrow \infty$	0.101
0.05	5.06×10^{-3}	2.83	0.101
0.1	5.07×10^{-3}	1.44	0.101
0.3	6.77×10^{-3}	1.013	0.025
1	2.04×10^{-2}	1.0002	0

(a) Decoupling costs work at every temperature, including in the entangled regime. (b) $W = k_B T \Delta I$ is recovered once $k_B T \gtrsim 0.3 \hbar \omega_0$. At $T = 0$, $k_B T I \rightarrow 0$ while the cost tends to the ground-state shift $\simeq \hbar \omega_0 q^2 / 8 \simeq \frac{1}{2} \hbar \omega_0 E_N^2$. (c) Entanglement vanishes above $k_B T_c \simeq \hbar \omega_0 / \ln(4/q)$. For the membrane experiment ($q = 4.25 \times 10^{-4}$, $\omega_0 / 2\pi = 400$ kHz) this gives $T_c \approx 2.1 \mu\text{K}$, against a room-temperature occupation of 1.6×10^7 phonons. A static cavity-mediated coupling therefore cannot entangle those membranes. Entanglement requires parametric (two-mode-squeezing) interactions and reservoir engineering, as in the experiments cited in the quantum note.

5 Mixed states: work tracks neither I nor E_N

For a two-mode squeezed thermal state (squeezing r applied to thermal occupation n per mode): $n_a = [(2n+1) \cosh 2r - 1]/2$, $I = 2g(n_a) - 2g(n)$, $E_N = \max\{0, 2r - \ln(2n+1)\}$. Ideal unsqueezing gives $W_{\text{on}} = -2\hbar\omega(2n+1) \sinh^2 r$. For $r = 0.5$:

n	I (nats)	E_N (nats)	W_{on} ($\hbar\omega$)
0	1.319	1.000	-0.543
0.5	0.922	0.307	-1.086
0.859	0.895	0	-1.476
5	0.869	0	-5.97
100	0.868	0	-109.2

Entanglement vanishes at $n = (e^{2r} - 1)/2 = 0.859$, and I saturates at $2 \ln \cosh 2r = 0.868$ nats, yet the released energy grows linearly in $2n+1$. The released energy is the state's ergotropy. Because the reduced states are thermal, and hence locally passive, all of it is attributable to correlations, but it is not a function of any correlation measure.[5, 6] What *is* fixed is the bath-assisted value of the correlations, $k_B T I$, from Eq. (4).

6 Implications for the two notes

Classical note: replace $q \approx \Lambda/\omega_0$ with $q \approx 2|\Lambda|/\omega_0$ and justify it via the normal-mode splitting (all I and W values $\times 4$); relabel the damping times as energy-decay times; correct the reference to 11, 4656; state that the experiment is a steady state, so $W = k_B T \Delta I$ applies only to a hypothetical equal-temperature version; add I_{ph} from Eq. (3) as the experimentally relevant dependence; and ask for (or digitize) $T_{1,2}^{\text{eff}}$, J , and Λ versus photon number. **Quantum note:** add the Eq. (4) explanation of the sign, the Gibbs-state table as the like-for-like counterpart, and the squeezed thermal family to show that I and E_N come apart and neither controls the work.

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